

FROM CODE TO CURES

**INTRODUCTION TO
BIOINFORMATICS &
COMPUTER-AIDED DRUG
DESIGN (CADD)**

*A Hands-on
Workshop for All AIU
Students*

 22 & 23 AUGUST 2026

 8:30 PM - 10:30 PM

 Online Workshop

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WORKSHOP SESSION | DAY 1

INTRODUCTION TO

BIOINFORMATICS &

COMPUTATIONAL BIOLOGY

Connecting Biology, Data Science, & AI

A comprehensive, hands-on workshop designed to build a strong, practical bridge from biological questions directly to state-of-the-art AI systems.

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WORKSHOP LEARNING OBJECTIVES.

01 Speak Biology

Use core molecular-biology vocabulary confidently: DNA, RNA, protein, gene, genome.

02 Locate the Fields

Explain how bioinformatics, computational biology, and data science differ and overlap.

03 Map Problems

Translate a biological question into a data science / ML problem statement.

04 Choose Methods

Match biological data types to the right ML/DL architecture (CNN, RNN, Transformer, GNN).

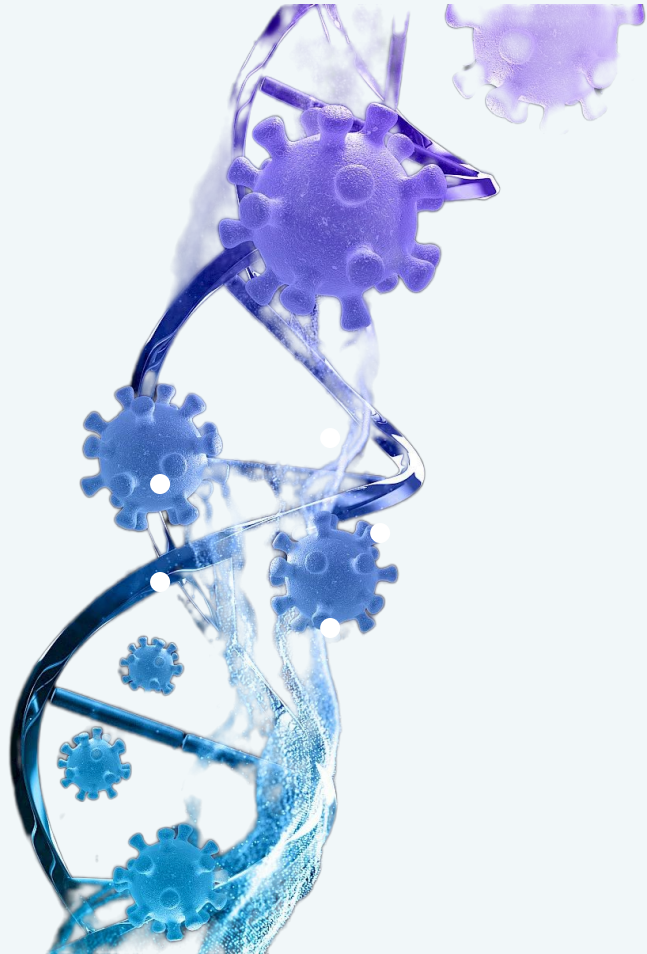
05 See the Frontier

Describe how generative AI and foundation models are reshaping biological discovery.

06 Start Building

Identify a concrete next step — a tool, dataset, or project — to begin contributing.

WHY CS STUDENTS SHOULD CARE.



01 Massive Scale

Biology has become a data-generation science: genomes, images, and signals at massive scale.

02 Algorithm-First

Every modern biological breakthrough leans on algorithms, not just wet-lab technique.

03 Direct Transfer

The tools you already know — Python, ML, statistics, systems thinking — transfer directly.

04 Emerging Careers

Emerging careers: computational biologist, bioinformatics engineer, AI-for-science researcher.

05 The Bridge

This workshop is the bridge: biological questions in, computational systems out.

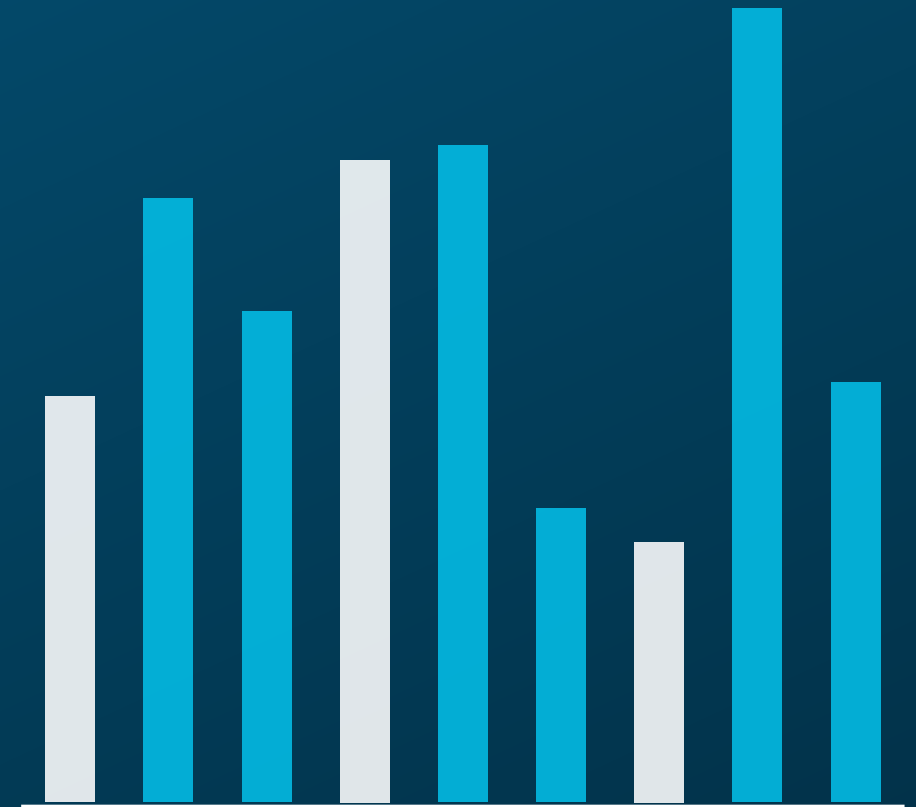
BIOLOGY AS A DATA SCIENCE.

~2 EB / YEAR

Projected Annual Genomic Data

Genomic sequencing alone is projected to generate more raw data per year than **astronomy**, **YouTube**, or **particle physics**.

This places biology among the largest data-generating domains on Earth, and squarely inside the territory of **data science and AI**.



BASIC MOLECULAR BIOLOGY

FOR CS STUDENTS.



01 Cell

The basic unit of life — think of it as a self-contained computer with local memory.

02 Nucleus

Stores the genome — analogous to persistent storage / the source-code repository.

03 Gene

A functional unit of DNA — like a callable function with inputs and outputs.

04 Protein

The 'executable' — genes are compiled (via RNA) into working molecular machines.

05 Mutation

A single-character edit to the source code, which may or may not change behavior.

DNA, RNA AND PROTEINS.

01

DNA

DNA stores hereditary information as a sequence over a 4-letter alphabet — A, C, G, T — a discrete, digital code copied every time a cell divides.

02

RNA

RNA is a working copy of DNA (mainly A, C, G, U) that carries instructions from the genome to the cell's protein-building machinery.

03

Proteins

Proteins are built from a 20-letter amino-acid alphabet and fold into 3D shapes that carry out almost every function in the cell — from catalysis to structure to signaling.

THE CENTRAL DOGMA.

01

DNA

Stores the hereditary blueprint inside the nucleus of every cell.

02

RNA

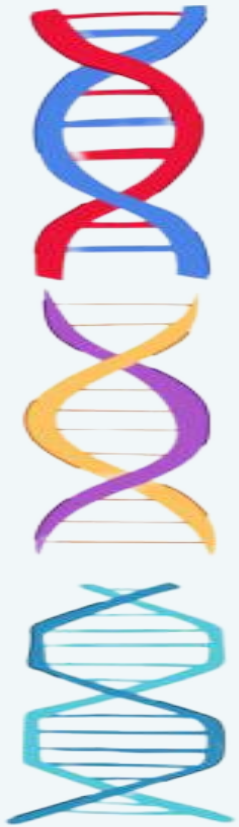
Transcription copies a gene from DNA into a messenger RNA (mRNA) strand.

03

Protein

Translation reads the mRNA at the ribosome and builds a folded protein.

GENES, GENOMES AND VARIATION.



01

Gene

A stretch of DNA that encodes a specific product, usually a protein.

02

Genome

The complete set of DNA in an organism — for humans, ~3.1 billion base pairs.

03

The Dark Genome

Only ~1–2% of the human genome directly codes for proteins; the rest regulates, structures, or remains poorly understood.

04

Variation

SNPs, insertions/deletions, and structural variants make every genome unique — the raw material for both evolution and disease.

WHERE BIOLOGICAL

DATA COMES FROM.



Sequencing

DNA/RNA read out as symbolic strings — the raw material of genomics.



Imaging

Microscopy, histopathology slides, and medical scans (MRI, CT, X-ray).



Mass Spectrometry

Identifies and quantifies proteins and metabolites in a sample.



Electronic Health Records

Structured + free-text clinical data linked to patient outcomes.



Structural Biology

X-ray crystallography, cryo-EM, and NMR resolve 3D molecular shapes.



Literature

Tens of millions of biomedical papers — a vast, underused text corpus.



PART TWO

WHAT IS

BIOINFORMATICS?

The computational discipline where biology meets code — turning raw biological sequences and measurements into searchable, comparable, and interpretable data.

01

Storing

Systematically archiving massive volumes of genomic and molecular datasets.

02

Retrieving

Building fast, indexed search systems to query complex biological records.

03

Organizing

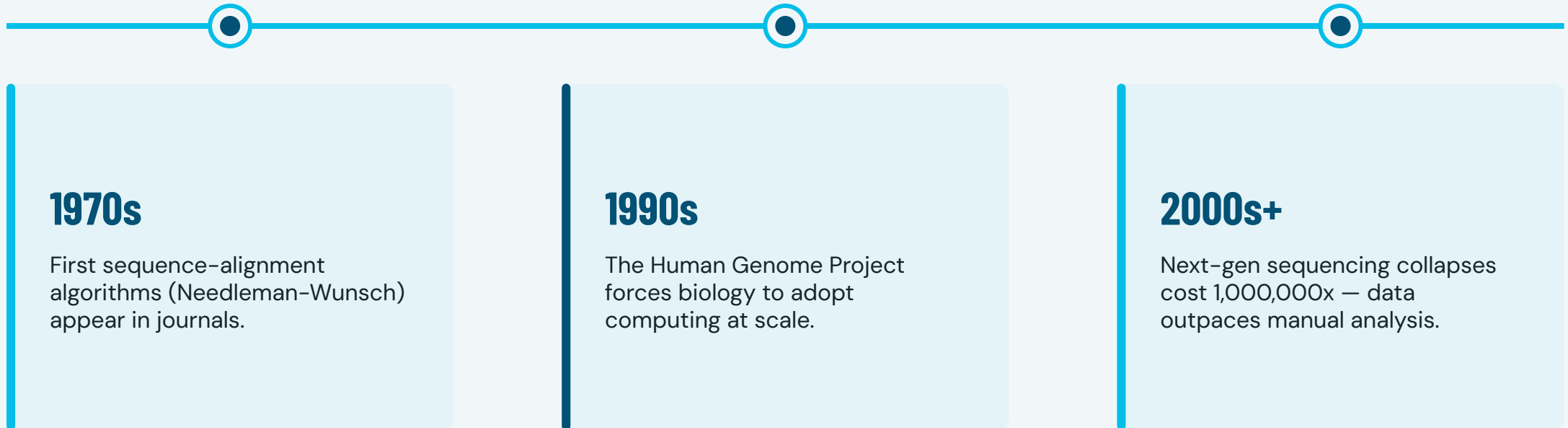
Structuring raw sequences and metadata into curated, standardized databases.

04

Analyzing

Applying mathematical and statistical models to extract meaningful insights.

WHY BIOINFORMATICS EMERGED.



THE INTERSECTION

THAT DEFINES IT.

BIOINFORMATICS

The ultimate convergence of computational power, mathematical modeling, and biological discovery to decode the complexity of life.

Molecular Biology

The biological foundation of genetic and molecular processes.

Computer Science

The infrastructure and systems used to process big data at scale.

Algorithms

Computational logic designed to align and analyze complex sequences.

Databases

Structured systems for archiving and querying biological records.

Mathematics

Mathematical formulas to model physical and biological systems.

Statistics

Statistical frameworks to distinguish signal from biological noise.

TYPES OF

BIOLOGICAL DATA.

Data Type	Example Format	Typical Size
Sequence	FASTA / FASTQ (DNA, RNA, protein)	KB – GB per sample
Structure	PDB coordinate files (3D atoms)	KB – MB per structure
Expression	Count matrices (genes x samples)	MB – GB per study
Image	Histopathology slides, MRI/CT scans	MB – GB per image
Clinical / Text	EHR records, biomedical literature	Structured + free text

BIOLOGICAL SEQUENCES:

DNA / RNA / PROTEIN.

DNA

Deoxyribonucleic Acid

Alphabet Size **4** (A,C,G,T)

Typical Length **$10^3 - 10^9$** bp

Stores / Does **Hereditary blueprint**

File Format **FASTA / FASTQ**

RNA

Ribonucleic Acid

Alphabet Size **4** (A,C,G,U)

Typical Length **$10^2 - 10^4$** nt

Stores / Does **Carries instructions**

File Format **FASTA / FASTQ**

Protein

Polypeptide Chain

Alphabet Size **20** amino acids

Typical Length **$10^1 - 10^4$** residues

Stores / Does **Performs function**

File Format **FASTA**

WHY WE NEED DATABASES.

- 01 Sequencing produces enormous volumes of raw, disconnected data every day, worldwide.
- 02 Databases give that data a stable identity: **accession numbers, versioning, cross-references**.
- 03 They enable comparison — *'is my new sequence similar to anything known?'* — at global scale.
- 04 Curated annotation turns raw sequences into **biological knowledge**: function, structure, disease links.
- 05 Public databases are the **training data source** for nearly every biological ML/DL model today.

CORE BIOLOGICAL DATABASES.



NCBI

US national hub for sequences, PubMed literature, and genome assemblies.



GenBank

Global, continuously updated public archive of nucleotide sequences.



UniProt

The reference database for protein sequence and function annotation.



PDB

Protein Data Bank — experimentally resolved 3D structures of biomolecules.



Ensembl

Genome browser and annotation resource spanning hundreds of species.



KEGG

Curated maps of metabolic and signaling pathways and interaction networks.

SEQUENCE

ALIGNMENT.

01 **Sequence Alignment** arranges two or more sequences to expose regions of similarity that may reflect shared ancestry, structure, or function.

02 **Pairwise alignment** (e.g. Needleman-Wunsch for global, Smith-Waterman for local) uses dynamic programming — the same algorithmic idea as edit distance in CS.

03 **Multiple sequence alignment** extends this to many sequences at once, forming the backbone of phylogenetics and conserved-motif discovery.



BLAST: BASIC LOCAL

ALIGNMENT SEARCH TOOL.

BLAST makes database-scale alignment tractable with a fast, heuristic three-step pipeline:

01

Seed

Find short exact matches



02

Extend

Grow matches outward



03

Score

Rank by significance (E-value)

GENOME

ANALYSIS.

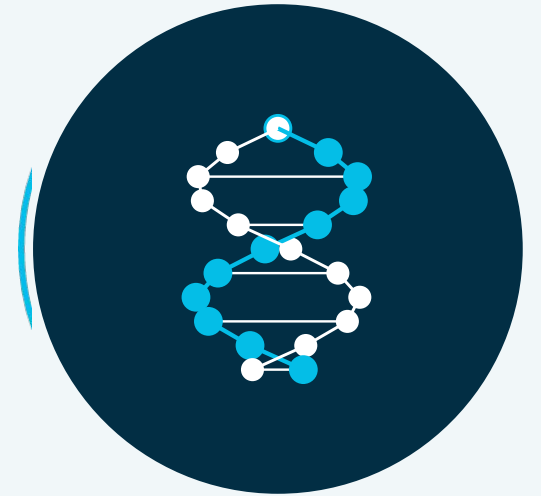
01 Assembly: reconstruct a full genome from millions of short, overlapping sequencing reads.

02 Alignment / mapping: place reads against a known reference genome.

03 Variant calling: identify positions where a genome differs from the reference.

04 Annotation: label genes, regulatory regions, and functional elements along the genome.

05 Comparative genomics: compare genomes across individuals or species to find patterns.



TRANSCRIPTOMICS & GENE EXPRESSION.

01

Measurement

Transcriptomics measures which genes are actively being transcribed into RNA, and how much, in a given cell or tissue at a given moment.



02

Technology

RNA-seq produces a matrix of genes × samples, where each value is an expression count — directly usable as an ML feature matrix.



03

Application

Expression patterns reveal cell identity, disease state, and treatment response, making transcriptomics a rich source of biological ML problems.

PROTEOMICS

STUDY.

01 

Overview

Proteomics studies the full set of proteins present in a cell, tissue, or organism — the 'executables' actually running at a given moment.

02 

Measurement

Mass spectrometry is the primary measurement tool: it identifies and quantifies proteins by their mass-to-charge signatures.

03 

Application

Unlike the genome, the proteome changes constantly with cell type, time, and condition — making it a dynamic, high-dimensional dataset well suited to ML analysis.

STRUCTURAL

BIOINFORMATICS.

Structural bioinformatics analyzes and predicts these levels computationally — the foundation for tools like AlphaFold.

01 

Primary

The linear amino-acid sequence itself

02 

Secondary

Local folding patterns: alpha-helices, beta-sheets

03 

Tertiary

The full 3D folded shape of a single protein chain

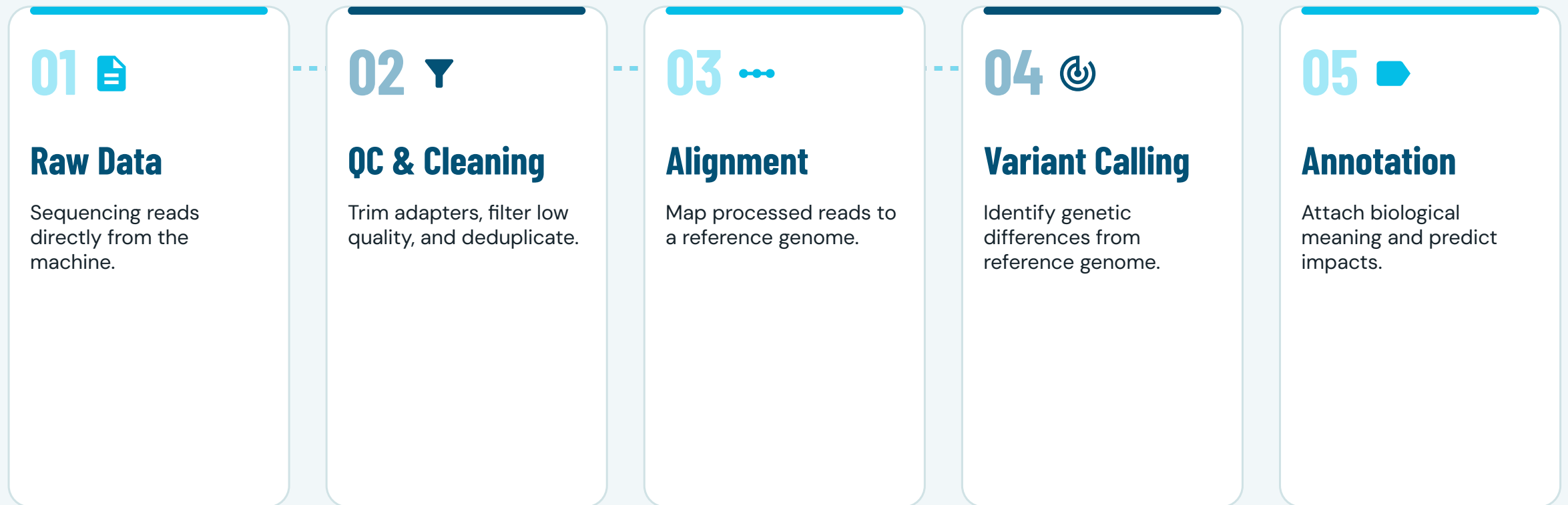
04 

Quaternary

Assembly of multiple protein chains into a complex

TYPICAL BIOINFORMATICS WORKFLOW.

A generic pipeline that turns raw sequencing output into biological insight — the template we will reuse across the whole workshop:



BIOINFORMATICS

GENERAL DISCUSSION

01

Key Topics

Vaccine Discoveries How bioinformatics accelerates vaccine breakthroughs like COVID-19 by replacing slow physical screening with fast computation.

Genetic Engineering Developing climate-resilient crops (e.g., modification of genetic structures to survive prolonged flooding conditions).

02

Core Applications

- **Molecular Medicine**
- **Personalized Medicine**
- **Preventative Medicine**
- **Gene Therapy**
- **Drug Development**

03

AI Integration

- **Target Discovery** AI screens millions of candidate molecules and accelerates trials via virtual predictions.
- **Custom Proteins** Generative models construct custom proteins for unique environmental conditions.
- **Precision Diagnostics** Integrates multimodal genomics and imaging data.

WHAT IS

COMP. BIOLOGY?

Computational biology uses mathematical, statistical, and computational models to understand how biological systems actually behave—moving beyond data storage to actively:

01 

Simulate

Create dynamic computational models to recreate and study the complex behaviors of biological systems.

02 

Predict

Forecast system responses and outcomes under mutations, drug interventions, or environmental shifts.

03 

Explain

Uncover the underlying physical and chemical mechanisms that govern life processes at a system scale.

Bioinformatics vs. Computational Biology

Bioinformatics

Focus

Managing & analyzing biological data

Core Question

How do we store, search, compare data?

Typical Output

Databases, alignments, annotations

Analogy

Data engineering for biology

Computational Biology

Focus

Modeling biological systems

Core Question

How does the system actually behave?

Typical Output

Simulations, mathematical models

Analogy

Applied math / systems modeling for biology

MODELING BIOLOGICAL SYSTEMS.

01

Differential-Equation

Models describe how molecule concentrations change over time.

02

Agent-Based

Models simulate individual cells or organisms interacting under simple rules.

03

Stochastic / Prob.

Models capture inherent randomness in molecular-scale events.

04

Graph-Based

Models represent genes, proteins, and metabolites as networks of interaction.

All of these models can be fit to data, simulated, and used to generate testable predictions.

BIOLOGICAL NETWORKS & SYSTEMS BIOLOGY.



Gene Regulatory Networks

Which genes switch other genes on or off, and when.



Protein-Protein Interaction

Which proteins physically bind or signal to each other.



Metabolic Networks

Chemical reaction pathways that convert nutrients into energy.



Signaling Networks

How cells relay external signals into internal responses.

MOLECULAR SIMULATION & PROTEIN STRUCTURE.

01 / Molecular Dynamics

Simulates the physical motion of atoms in a molecule over time, using physics-based force fields to model bonds, angles, and interactions.

02 / The Computational Bottleneck

These simulations are extremely expensive — a few microseconds of simulated protein motion can require weeks of supercomputer execution time.

03 / ML Surrogate Acceleration

This bottleneck is precisely why ML surrogate models are highly valuable: they approximate simulation-quality predictions in a fraction of the time.



MATH & STATISTICAL

MODELING IN BIOLOGY.



Differential Equations

Model continuous change: concentrations, population sizes.



Markov Chains / HMMs

Model sequential, state-dependent biological processes.



Bayesian Statistics

Combine prior knowledge with noisy experimental evidence.



Graph Theory

Analyze structure and flow within biological networks.

HOW COMP. BIOLOGY FRAMES A QUESTION.

01

Biological Question

e.g. 'why does this mutation cause disease?'

02

Formalize

Define variables, data, and assumptions

03

Model

Choose a mathematical / computational form

04

Solve

Simulate, optimize, or infer parameters

05

Interpret

Translate results back into biology

BIOLOGY MEETS DATA SCIENCE.



Once biological measurements exist as structured data — sequences, expression matrices, images — they become standard data science problems.

They are analyzed, modeled, and validated using exactly the same pipelines and rigorous workflows you already apply to any other dataset.

01 / Structured Data

Biological inputs formatted as features and samples:

- **Sequences**
DNA, RNA, and amino acid strings.
- **Matrices**
Gene expression and clinical tables.
- **Images**
High-resolution pathology & microscopy.

02 / The Pipeline

Standard logical stages applied to bio-datasets:

- **Cleaning & Prep**
Noise filtering and batch normalization.
- **Exploration**
Dimensionality reduction & statistics.
- **Modeling**
Machine learning and predictive inference.

A DATA SCIENCE PROBLEM.

01

Data Structure

Rows = samples (patients, cells, sequences); columns = features (genes, measurements).

02

Dimensionality

Biological data is often high-dimensional: thousands of genes measured on dozens of samples.

03

Data Quality

It is frequently noisy, batch-affected, and unevenly labeled — a realistic, messy dataset.

04

The Lifecycle

The same data science lifecycle applies: collect, clean, explore, model, validate, deploy.

FROM RAW DATA TO

MODEL-READY FEATURES.

01

Collection

Sequencing, imaging, and EHR data export.

02

Cleaning

Noise removal and handling missing clinical values.

03

Preprocessing

Data normalization, feature encoding, and format alignment.

04

Feature Engineering

Building robust, model-ready biological variables.

EXPLORATORY DATA ANALYSIS

FOR BIOLOGY.

01

Distribution Checks

Are expression values normally distributed, or heavily skewed?

02

Missingness Patterns

Are values missing at random, or systematically by batch/sample?

03

Correlation Structure

Which genes or features move together across samples?

04

Outlier Detection

Is a sample a biological signal, or a technical artifact?



The EDA Toolkit

Histograms, boxplots, correlation heatmaps, and pair plots provide the standard foundation for biological data exploration.

STATISTICAL ANALYSIS & HYPOTHESIS TESTING.

TESTING

Comparing Group Means

Is gene X expressed differently between disease and healthy samples? t-test / Mann-Whitney.

SIGNIFICANCE

The P-Value Concept

A p-value estimates how surprising the observed difference is under the 'no real effect' (null) hypothesis.

CORRECTION

Multiple-Testing Correction

Biological studies test thousands of genes at once. Correction is essential (Bonferroni, Benjamini-Hochberg / FDR) to control false positives.

RELEVANCE

Biological Relevance

Statistical significance is not the same as biological significance — always calculate and report effect size.

DIMENSIONALITY REDUCTION.

PCA

Principal Component Analysis

BEST FOR

Fast linear overview & noise reduction

PRESERVES

Global variance and overall structure

T-SNE

t-Stochastic Neighbor Embedding

BEST FOR

Visualizing local clusters and sub-populations

PRESERVES

Local neighborhoods and fine structures

UMAP

Uniform Manifold Approximation

BEST FOR

Large single-cell datasets, fast and scalable pipelines

PRESERVES

Both local and global data structures

CLUSTERING

BIOLOGICAL DATA.

K-MEANS

Feature-Based Grouping

Groups samples into a fixed number of clusters based on feature similarity metrics.

HIERARCHICAL

Nested Groupings

Builds a tree of nested groupings, which are directly related to phylogenetics and evolutionary history.

APPLICATIONS

Real-World Discovery

Discovering cell types from single-cell data, finding disease subtypes, and grouping co-expressed genes into functional modules.

VALIDITY

Biological Relevance

Cluster validity matters: biological functionality, not just mathematical distance, must confirm that the identified groups are real.

CLASSIFICATION & REGRESSION IN BIOLOGY.

CLASSIFICATION

Discrete Category Output

EXAMPLE BIOLOGICAL QUESTION

Is this tumor malignant or benign?

EXAMPLE BIOLOGICAL QUESTION

Does this sequence contain a promoter?

REGRESSION

Continuous Value Output

EXAMPLE BIOLOGICAL QUESTION

What is this patient's predicted survival time?

EXAMPLE BIOLOGICAL QUESTION

How strongly does this drug bind its target?

VISUALIZATION FOR DISCOVERY.



Heatmaps

Expression patterns across genes and samples at a glance.



Volcano Plots

Effect size vs. significance for thousands of genes at once.



PCA / UMAP Plots

Low-dimensional maps revealing sample or cell clusters.



Genome Browsers

Linear, zoomable views of annotated genomic regions.

THE COMPLETE DATA

SCIENCE PIPELINE.

01

Collect

Raw biological measurements



02

Clean

Handle noise & missing data



03

Explore

EDA, statistics, visualization

04

Reduce / Cluster

PCA, UMAP, k-means



05

Model

Classification / regression



06

Interpret

Translate results to biology

WHY MACHINE

LEARNING IS NEEDED.

The Challenge

LIMITS OF HAND-WRITTEN RULES

Biological patterns are often too high-dimensional, too subtle, or too nonlinear for manual formulas to capture.

- **High-Dimensional:** Thousands of features interact simultaneously.
- **Subtle & Complex:** Weak signals easily missed by static thresholds.
- **Rigid Formulas:** Coded rules fail to generalize to biological variation.

The Solution

DATA-DRIVEN PATTERN RECOGNITION

Machine learning lets us learn those complex patterns directly from biological data.

- **Direct Learning:** Models learn features directly from labeled data.
- **Adaptive Mapping:** Captures non-linear biological mechanisms.
- **No Hard-Coding:** Bypasses the need for rigid mathematical scoring.

TRADITIONAL VS.

ML-BASED BIOINFORMATICS.

TRADITIONAL BIOINFORMATICS

Rigid, theory-driven rules engineered manually by experts.

RULES

Hand-crafted (scoring matrices, thresholds)

GENERALIZATION

Limited to the coded rule set

DATA NEED

Works with small reference sets

EXAMPLE

BLAST alignment scoring

ML-BASED BIOINFORMATICS

Data-driven algorithms learning complex biological patterns.

RULES

Learned from labeled data

GENERALIZATION

Can generalize to new patterns

DATA NEED

Needs substantial training data

EXAMPLE

CNN-based variant classification

MAPPING BIOLOGY TO ML PROBLEMS.

Biological Problem		ML Problem Type
Disease diagnosis from a scan	→	Image classification
Gene function prediction	→	Sequence / multi-label classification
Drug-target binding affinity	→	Regression
Discover novel cell types	→	Unsupervised clustering
Protein structure prediction	→	Structured / sequence-to-structure prediction

CLASSIFICATION

EXAMPLES.



Cancer Diagnosis

Malignant vs. benign tumor from imaging or expression data.



Promoter Detection

Does this DNA region initiate transcription?



Protein Family

Which functional family does this protein sequence belong to?



Splice-Site Prediction

Is this genomic position a splice junction?

REGRESSION

EXAMPLES.



Drug Binding Affinity

Predict how strongly a molecule binds a target protein.



Gene Expression Level

Predict expression from upstream regulatory sequence.



Patient Survival Time

Predict time-to-event outcomes from clinical + genomic data.



Protein Stability

Predict the change in stability from a point mutation.

CLUSTERING

EXAMPLES.



Cell Type Discovery

Group single cells by expression similarity into novel types.



Disease Subtyping

Split a broad diagnosis into biologically distinct subtypes.



Co-expression Modules

Group genes that rise and fall together across conditions.



Sequence Family Discovery

Group unlabeled sequences into putative functional families.

FEATURE ENGINEERING FOR BIOLOGICAL DATA.



k-mer Counts

Frequency of every length-k substring in a DNA or protein sequence.



Summary Statistics

GC content, sequence length, and other simple biological summary statistics.



Domain Annotations

Known functional motifs or structural domains present within a sequence.



Expression-derived Features

Normalized counts, fold-changes, and downstream pathway activity scores.

LEARNED EMBEDDINGS: Increasingly replaced by learned embeddings from deep models (Part 6) — but still valuable for interpretability and small-data settings.

SEQUENCE

REPRESENTATION.



One-Hot Encoding

Each nucleotide/amino acid becomes a binary indicator vector.



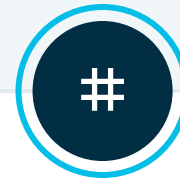
k-mer Frequency

Fixed-length vector of substring counts — simple and fast.



Learned Embeddings

Dense vectors from a trained sequence model (Part 6–7).

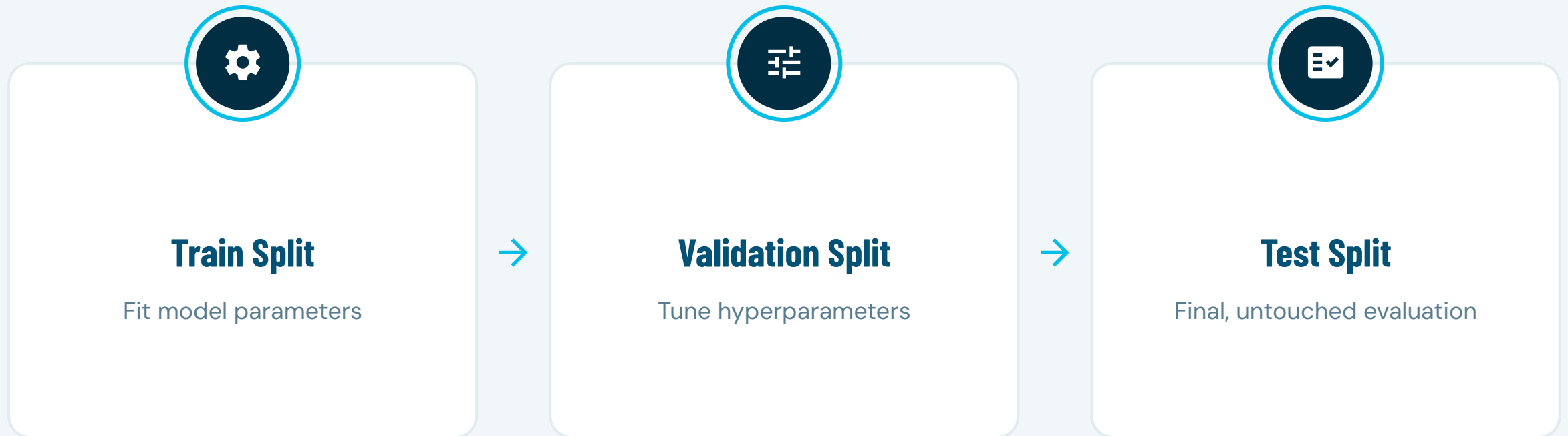


Positional Encoding

Explicitly represents where in the sequence a token sits.

TRAINING, VALIDATION & TESTING.

Biological data adds a wrinkle: splits must respect biological structure (e.g. no shared patients/genes across splits) to avoid leakage.



EVALUATION METRICS

FOR BIOLOGICAL ML.



Accuracy

Classes are roughly balanced



Precision / Recall

False positives and false negatives have different costs



ROC-AUC

Ranking quality matters, moderate imbalance



PR-AUC

Severe class imbalance (e.g. rare disease variants)



RMSE / MAE

Regression tasks (e.g. binding affinity, expression level)

COMMON PITFALLS IN BIOLOGICAL ML.



Class Imbalance

Rare positive cases (disease, mutation) get drowned out by negatives.



Small Data

Labeled biological data is expensive and slow to produce experimentally.



Data Leakage

Similar sequences/patients split across train and test inflate results.

EXPLAINABILITY & BIOLOGICAL INTERPRETATION.



Core Objective

A biological model's prediction is only useful if a scientist can act on WHY it was made.



Feature Importance

Feature importance (e.g. SHAP values) highlights which genes or positions drove a prediction.



Attention Weights

Attention weights in sequence models can point to biologically meaningful motifs or regions.



Collaborative Trust

Interpretability builds trust with wet-lab collaborators and guides follow-up experiments.

QUESTIONS & DISCUSSION.



01 // JOIN THE DISCUSSION

The Big Question

What biological problem would YOU want to point AI at?

Come up with a Research Question, I will guide you to find the Answer!



02 // GET IN TOUCH

WhatsApp

Connect directly for questions and networking.

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03 // RESOURCES & CODE

Workshop Materials

Slides, references, and starter code will be shared on GitHub:

[\[https://github.com/saky-semicolon\]](https://github.com/saky-semicolon)



References

1. Needleman, S. B., & Wunsch, C. D. (1970). A general method applicable to the search for similarities in the amino acid sequence of two proteins. *Journal of Molecular Biology*, 48(3), 443–453. [https://doi.org/10.1016/0022-2836\(70\)90057-4](https://doi.org/10.1016/0022-2836(70)90057-4)
2. Altschul, S. F., Gish, W., Miller, W., Myers, E. W., & Lipman, D. J. (1990). Basic local alignment search tool. *Journal of Molecular Biology*, 215(3), 403–410. [https://doi.org/10.1016/S0022-2836\(05\)80360-2](https://doi.org/10.1016/S0022-2836(05)80360-2)
3. International Human Genome Sequencing Consortium. (2001). Initial sequencing and analysis of the human genome. *Nature*, 409, 860–921. <https://doi.org/10.1038/35057062>
4. Venter, J. C., et al. (2001). The sequence of the human genome. *Science*, 291(5507), 1304–1351. <https://doi.org/10.1126/science.1058040>
5. Sayers, E. W., et al. (2022). Database resources of the National Center for Biotechnology Information. *Nucleic Acids Research*, 50(D1), D20–D26. <https://doi.org/10.1093/nar/gkab1112>
6. The UniProt Consortium. (2023). UniProt: The Universal Protein Knowledgebase in 2023. *Nucleic Acids Research*, 51(D1), D523–D531. <https://doi.org/10.1093/nar/gkac1052>
7. wwPDB Consortium. (2019). Protein Data Bank: The single global archive for 3D macromolecular structure data. *Nucleic Acids Research*, 47(D1), D520–D528. <https://doi.org/10.1093/nar/gky949>
8. Kanehisa, M., & Goto, S. (2000). KEGG: Kyoto Encyclopedia of Genes and Genomes. *Nucleic Acids Research*, 28(1), 27–30. <https://doi.org/10.1093/nar/28.1.27>
9. Jumper, J., Evans, R., Pritzel, A., et al. (2021). Highly accurate protein structure prediction with AlphaFold. *Nature*, 596(7873), 583–589. <https://doi.org/10.1038/s41586-021-03819-2>
10. Lin, Z., Akin, H., Rao, R., et al. (2023). Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637), 1123–1130. <https://doi.org/10.1126/science.ade2574>

